

## Customer Intelligence Analysis

**Project:** Customer Analytics Platform for Strategic Customer Retention and Revenue Growth

**Dataset:** Olist Brazilian E-commerce Dataset

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### 1. Executive Summary

This project develops a customer intelligence platform using the Olist e-commerce transactional dataset. The objective is to transform raw transactional data into a validated analytical framework for understanding customer behaviour, customer value, retention, revenue concentration, and customer experience.

The analysis covers 96,096 unique customers, 99,441 orders, 112,650 order items, 103,886 payment records, 32,951 products, 3,095 sellers, and 99,224 reviews.

A layered data architecture was implemented using Bronze, Silver and Gold layers. The data was profiled and subjected to 53 data quality checks covering uniqueness, referential integrity, domain validity, temporal consistency, missing values, and transaction-level integrity.

The analysis indicates that customer retention is a significant opportunity. 96.88% of customers made only one purchase, while only 3.12% became repeat customers. However, repeat customers generated a substantially higher average customer value of ₹314.99, compared with ₹161.82 for one-time customers.

Revenue is also highly concentrated. The highest-value 20% of customers account for 53.77% of total revenue. Most notably, the High-Value One-Time segment contains 36,055 customers, representing 37.52% of the customer base but contributing 68.16% of total revenue. These customers have demonstrated high willingness to spend but have not yet made a second purchase.

The analysis therefore suggests that the highest-priority customer strategy should combine high-value customer retention, second-purchase conversion, and customer-experience improvement.

### 2. Business Objective

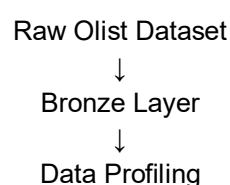
The primary objective is to answer the following business questions:

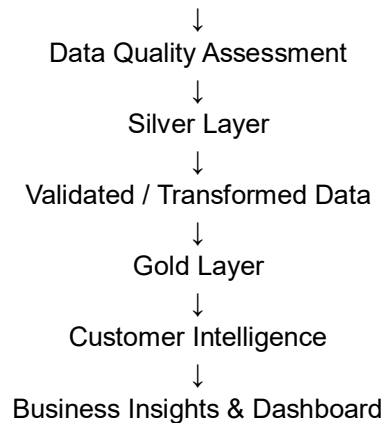
1. Who are the most valuable customers?
2. How effectively is the business converting customers into repeat purchasers?
3. Where is revenue concentrated across the customer base?
4. Which customer segments should receive different retention strategies?
5. Are customer experience problems concentrated in particular customer segments?
6. How can customer value and retention insights be translated into actionable business decisions?

### 3. Analytical Approach

#### 3.1 Data Architecture

The project follows a layered analytical architecture:





### **Bronze Layer**

The Bronze layer preserves the source data in structured PostgreSQL tables without applying analytical transformations.

The nine source entities are:

- Customers
- Orders
- Order Items
- Payments
- Products
- Sellers
- Reviews
- Geolocation
- Category Translation

### **Silver Layer**

The Silver layer applies cleaning, standardisation, business rules, and analytical transformations.

Key activities include:

- Standardising data types
- Handling missing values
- Deriving delivery metrics
- Establishing customer-level identifiers
- Aggregating transaction-level data
- Creating customer-level analytical features
- Validating transformed data

### **Gold Layer**

The Gold layer contains business-ready analytical datasets including:

- Customer 360
- Customer intelligence
- Customer experience
- Executive KPIs
- Customer segmentation
- Revenue concentration analysis

## **4. Data Quality & Reliability**

Before customer-level analysis, the dataset was subjected to 53 Data Quality Assessment checks.

The checks covered:

- Duplicate identifiers
- Missing identifiers
- Referential integrity
- Invalid monetary values
- Payment validity
- Review-score validity
- Temporal consistency
- Order-status consistency
- Product attributes
- Seller and geographic integrity
- Transaction completeness

The majority of structural and referential integrity checks passed.

Several findings required investigation rather than automatic correction. Examples included:

- Multiple customer\_id values associated with the same customer\_unique\_id
- Duplicate review IDs across different orders
- Timestamp sequencing anomalies
- Orders without associated items
- Missing product categories
- A small number of payment and product-attribute anomalies

Importantly, these findings were documented rather than silently removed.

The final analytical layer was subsequently validated to ensure:

- Unique customer grain
- Valid customer-level metrics
- Non-negative customer values
- Valid RFM scores
- Valid experience metrics
- Consistent customer segmentation

## 5. Customer Base Overview

The final analytical customer population contains:

| Metric                  | Result     |
|-------------------------|------------|
| Unique customers        | 96,096     |
| Total orders            | 99,441     |
| One-time customers      | 93,099     |
| Repeat customers        | 2,997      |
| Repeat customer rate    | 3.12%      |
| Total revenue           | ₹16.01M    |
| Average order value     | ₹160.99    |
| Average customer value  | ₹166.59    |
| Average delivery time   | 12.57 days |
| Average late-order rate | 6.81%      |

| Metric               | Result |
|----------------------|--------|
| Average review score | 4.08   |
| Poor-experience rate | 4.14%  |

The customer base is heavily skewed toward one-time purchasers, while revenue and customer value are concentrated among a smaller group of higher-value customers.

## 6. Business Findings

### 6.1 Customer Retention Represents a Major Growth Opportunity

#### Finding

The customer base is dominated by one-time purchasers.

- 93,099 customers are one-time customers.
- This represents 96.88% of the customer base.
- Only 2,997 customers, or 3.12%, made repeat purchases.

Despite their small population, repeat customers generate:

- ₹944,022.71 in revenue
- 5.90% of total revenue
- Average customer value of ₹314.99

This compares with an average customer value of ₹161.82 for one-time customers.

#### Interpretation

Repeat customers generate substantially greater cumulative value than one-time customers. Their advantage comes primarily from purchasing more frequently rather than placing larger individual orders.

Average orders:

- One-time customers: 1.00
- Repeat customers: 2.12

Average order value:

- One-time customers: ₹161.82
- Repeat customers: ₹148.85

#### Business Implication

The principal retention opportunity is therefore to increase purchase frequency and customer lifetime value, rather than simply attempting to increase the value of individual orders.

Potential interventions include:

- Second-purchase campaigns
- Personalised recommendations
- Cross-selling
- Loyalty programmes
- Post-purchase engagement
- Category-specific reactivation campaigns

### 6.2 Revenue Is Highly Concentrated Among High-Value Customers

Customer monetary value is highly skewed.

The highest monetary quintile contains approximately 20% of customers but contributes: 53.77% of total revenue

The lowest monetary quintile contributes only: 4.77% of total revenue

| Monetary Quintile | Customer Share | Revenue Share |
|-------------------|----------------|---------------|
| 1                 | 20%            | 4.77%         |
| 2                 | 20%            | 8.49%         |
| 3                 | 20%            | 13.06%        |
| 4                 | 20%            | 19.91%        |
| 5                 | 20%            | <b>53.77%</b> |

### Interpretation

Customer value is not evenly distributed across the customer base.

A uniform customer-management strategy would therefore allocate resources inefficiently.

### Business Implication

Retention and marketing resources should be allocated according to expected customer value.

High-value customers should receive more targeted retention and service interventions, while low-value inactive customers can be managed primarily through lower-cost automated campaigns.

## 6.3 High-Value One-Time Customers Are the Largest Identified Opportunity

This is the most significant customer-segmentation finding.

The High-Value One-Time segment contains:

| Metric                 | Result         |
|------------------------|----------------|
| Customers              | 36,055         |
| Customer share         | <b>37.52%</b>  |
| Revenue                | <b>₹10.91M</b> |
| Revenue share          | <b>68.16%</b>  |
| Average customer value | ₹302.63        |
| Average orders         | 1.00           |
| Average recency        | 287 days       |

These customers have generated substantial revenue despite having made only one purchase.

### Interpretation

This segment represents customers who have already demonstrated a high willingness to spend but have not yet transitioned into repeat purchasing.

This is a more actionable opportunity than simply targeting all one-time customers equally.

## Business Implication

A dedicated second-purchase conversion strategy should prioritise this segment.

Potential strategies include:

- Personalised product recommendations
- Cross-category recommendations
- Targeted second-purchase incentives
- Loyalty programme enrolment
- Follow-up campaigns based on purchase category
- Post-purchase engagement

The objective should be to convert a portion of these customers into repeat customers while protecting the value of their existing relationship.

### 6.4 High-Value Repeat Customers Are Valuable but Require Experience Protection

The High-Value Repeat segment contains:

- 2,383 customers
- Average customer value: **₹370.86**
- Average orders: **2.14**
- Average delivery time: **12.71 days**
- Average late-order rate: **6.18%**
- Average review score: **4.10**
- Poor-experience rate: **6.63%**

This is the highest-value customer segment on a per-customer basis.

However, its poor-experience rate is relatively high compared with several other segments.

## Interpretation

High-value repeat customers represent customers whose future value is particularly important. Service failures affecting this group therefore have potentially greater commercial consequences.

## Business Implication

The business should consider differentiated service monitoring for high-value repeat customers.

Potential actions include:

- Proactive delivery monitoring
- Delay notifications
- Service recovery
- Targeted loyalty benefits
- Priority customer support

The objective is to protect the lifetime value of customers who have already demonstrated repeat behaviour.

### 6.5 Delivery Performance Is Strongly Associated with Poor Customer Experience

The customer-experience analysis identifies a strong relationship between severe delivery problems and poor reviews.

Among the investigated poor-experience customers:

| Metric                  | Poor-Experience Customers |
|-------------------------|---------------------------|
| Average delivery time   | 37.41 days                |
| Average late-order rate | 100%                      |
| Average review score    | 1.13                      |
| Poor-experience rate    | 100%                      |

For customers without the poor-experience flag:

| Metric                  | Other Customers |
|-------------------------|-----------------|
| Average delivery time   | 12.55 days      |
| Average late-order rate | 0.03            |
| Average review score    | 4.14            |

### Interpretation

Customers experiencing severe delivery problems exhibit substantially poorer review outcomes.

This indicates that customer experience should not be analysed independently of fulfilment performance.

### Business Implication

Customer experience monitoring should integrate:

- Delivery duration
- Late-order rate
- Review scores
- Poor-experience flags
- Subsequent purchasing behaviour

This would allow the business to identify customers potentially at risk following a poor fulfilment experience.

### Analytical Caveat

The analysis demonstrates a strong **association**, not causality. Delivery delays may contribute to poor experiences, but the analysis does not establish that delivery delays alone cause lower customer satisfaction.

## 6.6 Recent One-Time Customers Represent an Earlier Intervention Opportunity

The Recent One-Time segment contains:

- 22,311 customers
- 23.22% of customers
- ₹1.63M in revenue
- 10.19% of total revenue
- Average recency: **139.9 days**
- Average customer value: **₹73.15**

Their average delivery time is approximately **10 days**, the lowest among the major one-time segments.

### Interpretation

These customers are comparatively recent purchasers and have not yet made a second purchase.

Unlike highly inactive customers, they may still be relatively accessible through timely post-purchase engagement.

### Business Implication

The business should intervene earlier in the customer lifecycle rather than waiting until customers become inactive.

Potential actions include:

- Post-purchase recommendations
- Product/category cross-selling
- Second-purchase reminders
- Loyalty programme invitations
- Time-sensitive personalised offers

### 6.7 Low-Value Inactive Customers Should Receive Lower-Cost Retention Interventions

The Low-Value Inactive segment contains:

- 15,755 customers
- 16.40% of customers
- ₹879,104.65 revenue
- 5.49% of total revenue
- Average customer value: **₹55.80**
- Average recency: **447.9 days**

### Interpretation

These customers have relatively low historical value and have been inactive for a substantial period.

The expected return from expensive individualised retention interventions is therefore likely to be lower than for high-value customers.

### Business Implication

This segment should primarily receive:

- Automated reactivation campaigns
- Low-cost promotional communication
- Broad category recommendations
- Periodic win-back campaigns

Marketing resources should be concentrated more heavily on higher-value customer segments.

## 7. Customer Segment Strategy

The segmentation framework translates the analysis into differentiated customer-management strategies.

| Customer Segment    | Strategic Priority | Recommended Approach                 |
|---------------------|--------------------|--------------------------------------|
| High-Value One-Time | Very High          | Convert to second purchase           |
| High-Value Repeat   | Very High          | Protect and retain                   |
| Developing Repeat   | High               | Encourage continued repeat behaviour |
| Recent One-Time     | High               | Early second-purchase activation     |



| Customer Segment     | Strategic Priority | Recommended Approach            |
|----------------------|--------------------|---------------------------------|
| Established One-Time | Medium             | Targeted reactivation           |
| Low-Value Inactive   | Low                | Low-cost automated reactivation |

This approach avoids treating all customers equally and instead aligns customer-management effort with expected economic value and behavioural opportunity.

## 8. Business Recommendations

### Recommendation 1 — Develop a Value-Based Retention Programme

Customer engagement should be differentiated according to:

#### Customer Value + Recency + Purchase Frequency

Rather than treating the 96,096 customers as one homogeneous population, the business should use the customer segments to determine retention priorities.

### Recommendation 2 — Prioritise High-Value One-Time Customers

The High-Value One-Time segment represents the largest identifiable opportunity.

With:

#### 37.52% of customers generating 68.16% of revenue

this segment should receive dedicated second-purchase conversion initiatives.

Success should be measured through:

- Second-purchase conversion rate
- Repeat customer rate
- Incremental revenue
- Customer value after conversion

### Recommendation 3 — Protect High-Value Repeat Customers

High-value repeat customers already demonstrate strong customer economics.

Management should therefore prioritise:

- Service reliability
- Delivery monitoring
- Customer recovery
- Loyalty incentives

The objective is to prevent high-value customers from becoming inactive.

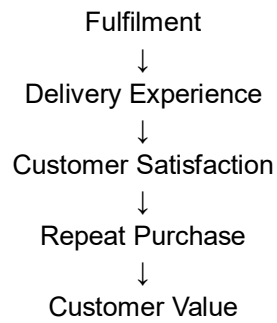
### Recommendation 4 — Integrate Customer and Operational Analytics

Customer analytics should incorporate operational indicators such as:

- Delivery duration
- Late-order rate
- Review score
- Poor-experience rate

This creates an integrated view of the customer journey:

Order  
↓



This framework can support future churn and retention modelling.

## 9. Executive KPI Summary

| KPI                                 | Result     |
|-------------------------------------|------------|
| Total customers                     | 96,096     |
| Total orders                        | 99,441     |
| Total revenue                       | ₹16.01M    |
| Repeat customer rate                | 3.12%      |
| Average order value                 | ₹160.99    |
| Average customer value              | ₹166.59    |
| Repeat customer value               | ₹314.99    |
| Repeat customer revenue share       | 5.90%      |
| Average delivery time               | 12.57 days |
| Average late-order rate             | 6.81%      |
| Average review score                | 4.08       |
| Poor-experience rate                | 4.14%      |
| Top monetary quintile revenue share | 53.77%     |
| High-Value One-Time revenue share   | 68.16%     |

## 10. Limitations

The analysis has several limitations that should be considered when interpreting the results.

### 10.1 Historical Dataset

The analysis is based on historical Olist transactions and therefore describes observed customer behaviour rather than current customer behaviour.

### 10.2 Customer Identification

The analysis uses customer\_unique\_id as the customer-level identifier. The distinction between customer\_id and customer\_unique\_id means that customer-account and underlying-customer concepts should not be treated as interchangeable.

### 10.3 Observational Analysis

The findings identify patterns and associations but do not establish causal relationships.

For example, the association between delivery delays and poor reviews does not by itself prove that delivery delays caused the negative reviews.

#### 10.4 No Marketing Exposure Data

The dataset does not contain detailed information on:

- Marketing campaigns
- Discounts received
- Customer acquisition cost
- Email engagement
- Advertising exposure

Therefore, the analysis cannot directly estimate campaign-level return on investment.

#### 10.5 No Predictive Model

The current project is primarily descriptive and diagnostic.

A future extension could develop predictive models for:

- Repeat-purchase probability
- Customer churn
- Customer lifetime value
- Poor-experience risk

### 11. Future Analytical Extensions

The current customer intelligence platform provides a foundation for more advanced analytics.

Potential extensions include:

- **Repeat-Purchase Prediction-** Predict the probability that a one-time customer will make another purchase.
- **Customer Churn Prediction-** Identify customers at risk of becoming inactive.
- **Customer Lifetime Value-** Estimate expected future customer revenue rather than relying solely on historical monetary value.
- **Recommendation Engine-** Use customer purchase history and product/category behaviour to generate personalised product recommendations.
- **Marketing Uplift Modelling-** If campaign-treatment data becomes available, estimate which customers are most likely to respond to specific interventions.
- **Cohort Analysis-** Track retention and revenue behaviour based on customer acquisition/order cohorts.

### 12. Conclusion

The analysis demonstrates that the Olist transactional dataset can be transformed into a customer intelligence framework capable of supporting customer segmentation, retention analysis, revenue concentration analysis, and customer-experience management.

The most important finding is that **customer value is highly concentrated while repeat purchasing remains limited**. Only 3.12% of customers are repeat customers, yet repeat customers have nearly twice the average customer value of one-time customers.

The **High-Value One-Time** segment presents the clearest commercial opportunity. These customers represent 37.52% of the customer base and 68.16% of revenue while having made only one purchase.

At the same time, high-value repeat customers require protection because their economic contribution makes service failures potentially more consequential.

Overall, the analysis supports a shift from **uniform customer management toward value-based customer management**, where retention investment, customer experience interventions, and marketing activity are prioritised according to customer value, recency, frequency, and observed experience.

#### **Appendix A — Analytical Layer Outputs**

The final Gold layer contains business-ready datasets supporting the analysis:

| <b>Gold Dataset</b>        | <b>Purpose</b>                                      |
|----------------------------|-----------------------------------------------------|
| customer_360               | Unified customer-level analytical view              |
| customer_intelligence      | Customer value, frequency, recency and segmentation |
| customer_experience        | Delivery, lateness, reviews and experience          |
| executive_kpis             | Business-level KPI summary                          |
| Customer segment analysis  | Segment-level revenue and customer economics        |
| Monetary quintile analysis | Revenue concentration                               |
| Customer type analysis     | One-time vs repeat customer comparison              |